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Chapter 2

The persuasiveness of persuasive discourse

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Abstract

This investigation explored why and how persuasion occurs. Toward this end, we examined the processing of two articles under conditions called for in the persuasion and conceptual change literatures. One unique aspect of the current study was the use of topic-specific measures of beliefs, knowledge (i.e., perceived and demonstrated), and interest. The performance of members of three educational communities (i.e., undergraduates, graduate students, and faculty) was compared, as was the knowledge and interest of more or less persuaded readers. One article argued that there are prenatal similarities among humans and other mammals, while the second presented a favorable view of same-sex marriages. Although both articles produced significant changes in readers' beliefs, knowledge, and interest, there were performance differences by text, educational community, and persuasion group. Results showed that readers' perceptions of their knowledge played a more significant role in persuasion outcomes than the knowledge they actually demonstrated. Outcomes suggest the need to look more closely at the relationship between knowledge and interest in subsequent studies of persuasion. © 2002 Elsevier Science Ltd. All rights reserved.

The goal of this research was to extend current understanding of the conditions, which foster changes in readers' knowledge and beliefs about contestable issues. Most individuals have encountered some written or verbal message that has markedly altered their views or deeply touched their emotions. Such a message may have stimulated their thinking in surprising and unexpected ways. It may have strengthened their position or resolve on a given topic, caused them to react with anger or disbelief, or led them to see some sensitive or controversial subject in a positive light. The message may even have been powerful enough to provoke them to act or to shift their perspective on the stated issues. That is to say, they may have found themselves *persuaded* (Miller, 1980; Stiff, 1994). As mature learners, we have

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all been persuaded at some point. The fact that persuasion occurs is, thus, unquestionable. Why and how persuasion occurs is an entirely different matter. Indeed, such questions have been the catalyst for centuries of philosophical debate and psychological inquiry.

A fascination with persuasion reaches deep into human history, as evidenced by the dialogues of Plato (Carey, 1994) and the treatises of Aristotle (Cooper, 1932). In ancient times, oration was the art of verbal persuasion, and Plato, in his dialogues, raised doubts about the divinity of oration. In effect, Plato was among the first to claim that oration, as a persuasive art, could be learned. Aristotle, Plato's pupil, forwarded one of the earliest models of persuasion, declaring that the keys to persuasion lie in the *ethos* (i.e., objective characteristics) of the author, the *pathos* (i.e., subjectivity or emotionalism) of the listener, and in the arguments themselves. Even though current theories and models of persuasion and conceptual change have become increasingly more elaborate and sophisticated, they still pay homage to the ideas forwarded by Plato and Aristotle.

For instance, rhetoricians and philosophers stress how aspects of the message can stimulate or stifle the persuasion process (Chambliss, 1994, 1995; Toulmin, 1958). Therefore, in judging the persuasiveness of a message, they inquire whether the author's premises are clearly stated, the arguments and counter-arguments well developed, and whether the author evokes the emotions of the audience. Similarly, social psychologists frequently explore how various aspects of the source, message, recipient, and media influence attitude change (Hovland, Janis, & Kelley, 1953; Jaccard, 1981; Johnson, 1994). Such researchers ask whether the reader or listener finds the author credible, has experiences relative to the topic, or can make sense of the message. As with others (Dole & Sinatra, 1994, 1999; Petty & Cacioppo, 1981, 1986), we hold that persuasion represents a confluence of such person and message factors. That is, while characteristics of the person and the message can be statistically isolated and analyzed, it is truly their interplay that affords the greatest understanding as to why and how persuasion occurs.

Various models of persuasion (e.g., Elaboration Likelihood Model or ELM; Petty & Cacioppo, 1986) and conceptual change (e.g., Cognitive Reconstruction of Knowledge Model or CRKM; Dole & Sinatra, 1998) acknowledge that changes in attitude or conceptual understanding occur as a result of the interaction of person and text. For example, the ELM, the best known and most tested model of persuasion (Stiff, 1994), proposes that a persuasive message can be processed through one of two routes, the central or the peripheral. The central processing route is pursued when there is thoughtful consideration of a message or its arguments. This route is presumed to produce changes in schematic structures, thereby resulting in enduring shifts in attitude. In contrast, peripheral processing results from attention to simple cues contained in the persuasive message or context (e.g., page length). Thus, this processing route can result in relatively temporary attitude changes.

The specific route pursued and the degree to which this path results in a marked change of attitude depends on several factors. For example, the quality of the stated arguments (i.e., strong and weak) affect individuals' responses to the message. Additionally, peripheral cues not directly associated with the message or its

arguments can also influence outcomes. For instance, an overly hot room or a particularly long reading can spark attitude change because these qualities trigger rather primitive affective states. From the central processing route, elaborations (i.e., the scrutiny of the stated arguments) also influence persuasion and are a consequence of individuals' motivation and their ability to process the message.

Similarly, the CRKM describes learner (e.g., existing conceptions or motivation) and text (e.g., comprehensibility or coherence) conditions that result in the restructuring of conceptual knowledge by eliciting learner engagement. It is through such engagement that concepts are restructured. The CRKM posits three qualities as highly predictive of conceptual change: strength of existing conceptions, conceptual coherence, and individuals' commitment or need to believe. Another learner variable specified in this model is the construct of motivation or a person's desire to process new concepts. The CRKM, like the ELM, proposes that motivation encompasses personal responsibility and a "need for cognition," as well as conceptual dissatisfaction (Chinn & Brewer, 1993; West & Pines, 1985) and the social context (Guzzetti & Hynd, 1998).

1. Conditions central to learning and development

Even though the ELM and CRKM address the interplay of person and text factors, these models do not address some of the fundamental conditions that we (e.g., Alexander, 1998; Murphy & Woods, 1996) and others (e.g., Pintrich, 1994) have identified as central to learning and development. In particular, our concerns center on:

- the specificity of the constructs identified in these models;
- the potential for interplay among person variables;
- the influence of processing non-contrived, controversial texts;
- intra-individual response patterns; and
- the impact of respondents' educational backgrounds.

1.1. Construct specificity

In our research, we have determined that a focus on very broad variables, such as those indicated in the ELM and CRKM (e.g., prior knowledge and motivation), can mask significant effects in learning and development (Alexander, Jetton, & Kulikowich, 1995; Alexander, Kulikowich, & Schulze, 1994; Alexander & Murphy, 1998). In fact, Dole and Sinatra (1998) explicitly note that there are multiple forms of knowledge that need to be considered in conceptual change. Thus, in the current study, the general construct of prior knowledge was replaced with two more specific knowledge forms, perceived (i.e., self-reported) knowledge and demonstrated (i.e., tested) knowledge.

The decision to compare findings for both perceived and demonstrated knowledge was sparked by the intriguing results from an earlier study (Murphy, 1998).

Specifically, using both self-report and open-ended measures of knowledge, Murphy found that readers' perceptions of their knowledge were more powerful predictors of persuasion than the knowledge they actually demonstrated. For that reason, similar self-report and open-ended measures were used in the current study.

Like the term *prior knowledge*, motivation encompasses a plethora of important, related constructs in the literature (Murphy & Alexander, 2000; Pintrich, 1994). Because persuasion entails the response to a *particular* message forwarding *specific* arguments about a *certain* topic, we think topic-specific interest influences this change process. By topic-specific interest, we mean the level of interest that individuals report for a particular subject, such as the legalization of same-sex marriages. Interest is mentioned in both the ELM and the CRKM, but as a dimension of motivation dealing with general relevance. Further, in these models, higher levels of interest translate into greater changes in attitudes or conceptual understanding. Yet, our own research would suggest a less linear relationship between topic interest and persuasion, a point elaborated in the subsequent section on interplay (Murphy & Alexander, 2001).

1.2. *Interplay of person variables*

In their model, Petty and Cacioppo (1986) state that higher levels of prior knowledge can translate into biased processing of textual arguments, thus negatively affecting persuasion. The results from our exploratory study of persuasion support this knowledge pattern (Alexander, Murphy, Buehl, & Sperl, 1998). In essence, a threshold of knowledge was required for persuasion to occur. Too little knowledge appeared to result in peripheral processing or limited engagement of the text, and too much knowledge seemed to produce the type of biased processing the ELM and CRKM would predict.

Because knowledge of the topic and interest in that topic have been inextricably intertwined in our research (e.g., Alexander & Dochy, 1995; Alexander & Murphy, 1998), a similar threshold can be predicted for interest as for knowledge. That is, when interest in the topic is too low there is superficial or limited processing of the controversial message. When interest in the topic is very high, however, we would assume that the individuals might come to that message with biases that restrict their ability to benefit from the authors' arguments.

1.3. *Naturally occurring texts and controversial issues*

Much of the research on persuasion centers on discourse expressly crafted to highlight particular textual features or elements, such as the argument structure or cohesion. As a result, these numerous investigations and the predictions they instigate may not generalize to the unmodified discourse found in individuals' everyday environments. For example, in a number of studies, textual materials were either "fabricated" (Slusher & Anderson, 1996, p. 113) or crafted from one source (e.g., an article from *Science*; Kardash & Scholes, 1995) or multiple sources (e.g., newspapers and public health information; Kardash & Scholes, 1996). Certainly, in

the case of the ELM (Petty & Cacioppo, 1986), most of the authors' studies employed short texts, one or two paragraphs in length. In addition, the experimental materials were often contrived by the researchers to highlight the argument structure.

Petty and Cacioppo (1986) explicitly state that it would be difficult to assess persuasion under more natural conditions, as with the reading of unmodified or uncontrived texts. Yet, reading naturalistic, persuasive texts is one aspect of ecological validity expressly called for in the literature (Bronfenbrenner, 1989; Chambliss & Garner, 1996). Consequently, naturally occurring texts were used in this study rather than messages contrived or modified for experimental purposes. The phrase *naturally occurring* refers to persuasive materials that likely exist in individuals' environments (e.g., newspaper editorials, magazine articles, or pamphlets) and which are presented to readers without modification. In this study, participants read two articles from *Life* in their entirety and directly from the magazine. The use of non-contrived texts afforded us the opportunity to extend the current research on persuasion that has relied heavily on specially constructed texts.

Beyond this issue of contrived versus naturally occurring texts, the role that the underlying topic played in the persuasive process was also of interest. In this investigation, participants read controversial texts about prenatal differences between humans and other mammals, and about the legalization of same-sex marriages. Although other researchers have targeted controversial subjects, they have not always viewed persuasion as a shift in general stance relative to a contentious idea. Instead, they have focused on changes in readers' understanding of particular and verifiable content associated with that topic.

For example, Kardash and Scholes (1995, 1996) and Slusher and Anderson (1996) examined belief change about AIDS by gauging readers' positions on a series of verifiable statements, such as the idea that AIDS can be transmitted by casual contact or by mosquito bites. They described individuals' views on these statements prior to and following their reading of informational text as either "accurate" or "inaccurate" beliefs. Even though we agree that the accuracy and inaccuracy of individuals' notions can be important aspects of the persuasion process, we would place these more factual understandings under the heading of *knowledge* in the current investigation. A more encompassing orientation toward a controversial topic would be indicative of individuals' *beliefs* (e.g., same sex marriages *should* or *should not* be legalized)—an orientation that is not simply a reflection of the depth or accuracy of knowledge (Chinn & Brewer, 1993).

1.4. Intra-individual explorations

Although empirical research on persuasion and conceptual change has been extensive, explorations of intra-individual differences have been relatively rare. This issue is important from the perspective of generalizability. Most often, in the literature, findings regarding the persuasion process are based on a single text and, thus, a single topic. Under such conditions, the question of generalizability of findings cannot be adequately addressed. To examine the consistency of identified

patterns across situations, each participant in this study was asked to read and respond to two controversial articles. Participants' beliefs, knowledge, and interests relative to those articles were measured both prior to and following reading. In this way, we hoped to increase the generalizability of our findings.

Along with intelligence, Petty and Cacioppo (1986) identify educational level as an important aspect of the ability to process. However, they go on to state that because of the sparse literature on the role of educational level, they could not consider this variable in any detailed way in the ELM. We have found it invaluable to explore knowledge and beliefs from the standpoint of participants' educational level (e.g., Alexander & Dochy, 1995; Alexander, Murphy, Guan, & Murphy, 1998; Alexander, Murphy, & Woods, 1997). In these studies, we ascertained that individuals in varying educational communities (e.g., high school students, undergraduates, and experts) displayed distinct patterns of beliefs, knowledge, and interests relative to an array of topics. However, we had not considered how the distinctions among these educational communities might influence the processing of controversial texts.

Based on the aforementioned issues, the following questions guided this study:

- What changes occur in participants' beliefs, perceived knowledge, demonstrated knowledge, and topic-specific interests as a consequence of reading texts that forward a particular view on a controversial topic?
- In what ways do mature readers respond differentially to two naturally occurring texts that present certain views on controversial topics?
- How does the educational level of mature readers influence the nature and degree to which they are persuaded by written text?
- How do the profiles of those who are more persuaded by their reading of controversial texts differ from the profiles of those who are less persuaded?

2. Method

As noted, participants in this study represented three educational communities—undergraduate students, graduate students, and faculty. Specifically, participants included 37 undergraduate students, 25 graduate students, and 20 faculty from a large, urban, land-grant university in the eastern United States. The 37 undergraduates, mostly juniors and seniors, were enrolled in an upper level educational psychology course. These undergraduates were ethnically diverse (approximately 51% Caucasian, 19% African American, 14% Hispanic, 8% Asian, and 8% mixed or unspecified) and represented a variety of majors (e.g., psychology, criminal justice, physical education, and business). The breakdown of females and males was relatively consistent with undergraduate education courses with 26 females (70.3%) and 11 males (29.7%).

Twenty-five masters and doctoral students from the same institution also took part in this study. Most of these individuals were pursuing degrees in human development, although other majors (e.g., counseling and student services) were represented. The breakdown of females and males was 20 (80%) and 5 (20%),

respectively. Twenty-three of these respondents were Caucasian and two were African American.

Twenty faculty members from colleges of education also volunteered to be participants in this investigation. Faculty who took part in the study are nationally recognized experts in human development, educational psychology, special education, teacher education, measurement and statistics, and literacy. In addition, several have been involved in research on persuasion. There were 11 females (55%) and 9 males (45%) among the faculty respondents. Nineteen of these respondents were Caucasian and one was African American.

2.1. Texts

For the experimental texts, two articles were selected from the same issue of *Life* magazine. Prior research had shown such professionally crafted articles to be suitable in length, reading difficulty, and appeal for our targeted communities (Buehl, Alexander, Murphy, & Sperl, 2001; Murphy & Alexander, 2001). These articles were two-sided refutational texts, which means that the authors present both sides of the issue although one side is more strongly and positively supported while the other is rebutted. One of the articles, *What does it mean to be one of us?* (Miller, 1996), made the case for similarities in the embryos and fetuses of humans and other animals (e.g., chickens, pigs, and chimpanzees). This article, referred to as the Prenatal Similarities text, consisted of 15 pages and 2955 words, and incorporated a captivating display of 25 photographs to support its contention of similarities, along with extensive factual information.

The second article, *When we talk about gay marriage, what do we really mean?* (Ferrato, Dowling, & Nyary, 1996), took a favorable stance on the legalization of same-sex marriages. This article, referred to as the Same-Sex Marriage text, was 8 pages long and contained 1171 words and 8 photographs. Unlike the Prenatal Similarities text, the Same-Sex Marriage text included limited factual content, relying instead on personal information about a gay couple trying to deal with the lack of legal support for their non-traditional lifestyle. Participants read these articles directly from the magazine in counterbalanced order. We believed that any attempt to copy these articles or remove them from the magazine would alter their appearance and decrease the ecological validity of the study.

2.2. Pre-reading response task

Participants were asked a series of topic-specific questions before they read the articles. These questions provided information about respondents' initial beliefs relative to each article's stated premise, their perceived knowledge, their demonstrated knowledge, and their interest in the topic.

2.2.1. Topic beliefs

Participants indicated their stance on a statement derived from the main premise of each article. As illustrated below for the Same-Sex Marriage article, each

participant indicated his or her position on the statement by placing an X on a line (150 mm) running from *strongly disagree* to *strongly agree*. This variation of a Likert scale allows resulting data to be analyzed in a continuous rather than discrete manner.

The federal government should move to legalize same-sex marriages.

strongly
disagree

strongly
agree

Responses were recorded in millimeters, with higher numbers indicative of a stronger agreement with the authors' premise. To decrease the effect for social desirability, the position statement for the Prenatal Similarities article was worded counter to the actual premise of the article. Specifically, we asked participants to indicate their view on the statement: "Before birth, humans are uniquely different from all other forms of animal life." Responses to this statement were reverse coded so that higher numbers signified greater agreement with the author's actual claim (i.e., that there are great similarities prenatally between human embryos and fetuses and those of other animals).

2.2.2. Topic knowledge

Participants' knowledge of the article topics was gauged in two ways. First, their perceived knowledge was assessed by asking them "to indicate how much you think you know" about prenatal differences or the legalization of same-sex marriages on a continuum from *relatively nothing* to *a great deal*. Once again a 150 mm line was used for this assessment and scored it in a fashion similar to the previous two questions. Second, as a measure of *demonstrated knowledge*, participants were asked to share their knowledge about each topic. Specifically, prior to reading, participants were asked: "What background information or knowledge helped you form your position on this issue [prenatal differences or legalization of same-sex marriages]? In other words, tell us what you already know about this subject." Participants' responses were scored by marking and categorizing each information unit in their answers in a manner described in the section on coding and reliability. For this scoring, an information unit was operationally defined as a complete and accurate thought, roughly equivalent to a phrase or independent clause.

2.2.3. Topic interest

In a similar fashion, participants indicated their interest in the article's topic on a continuum from *very disinterested* to *very interested*. Specifically, they were asked to indicate how interested they were in reading about prenatal similarities and differences between humans and other animals or about the legalization of same-sex marriages. Responses were scored on the 150 mm line in a manner identical to the beliefs question.

2.2.4. *Post-reading response task*

Immediately after reading each article, participants responded to a set of questions similar to those included in the pre-reading task. The belief statement and the directions to respondents were identical to those at pre-reading. With regard to topic knowledge and interest assessments, variations of the pre-reading questions were used. For example, for topic interest participants were asked to indicate “how interesting you found this topic [prenatal differences or legalization of same-sex marriages] to be.” Similarly, when marking their level of perceived knowledge, respondents were directed to “indicate how much you think you now know about” prenatal differences between humans and other animals, or the legalization of same-sex marriages.

This same approach to assessing readers’ beliefs, perceived knowledge, and interest had been used in a prior study (Buehl et al., 2001) with strong test-retest reliabilities (i.e., 0.74–0.97). Finally, the question assessing participants’ demonstrated knowledge read as follows: “What specifically did you learn from reading this article about [prenatal differences or legalization of same-sex marriages]? In other words, what do you remember from this article?”

2.3. *Coding and reliabilities*

Three doctoral students were taught the rubric for coding participants’ responses to the demonstrated knowledge question. Responses were coded based on the number of information units offered by participants. These information units were basically statements containing factual information related to each article (e.g., “Hawaii is trying to legalize gay marriages”). Because of the distinction in wording for the pre-reading and the post-reading demonstrated knowledge questions (i.e., “tell what you already know” versus “tell me what learned from the article”), these two items were scored differently. Specifically, pre-reading demonstrated knowledge was a simple total of all information units embodied in participants’ written responses to the pre-reading question. Post-reading demonstrated knowledge, in contrast, was the sum of any new information units contained in responses to the post-reading question plus the number of information units in the pre-reading task. Our decision to score demonstrated knowledge in this manner was based on the directive to participants to tell us what they learned from the reading these articles.

Information units were not the only content in participants’ written responses to the demonstrated knowledge question, however. Although not included in our final analysis, we also noted contextual and evaluative statements. Contextual statements conveyed information of a more personal nature, such as participants’ background or experiences (e.g., “I went to Catholic school” or “I am gay”), or they described where or from whom they learned the information that they included in their responses (e.g., “I have gay friends who talk about this issue” or “The article specifically said that...”). Evaluative statements were marker clauses or phrases that indicated how the participant judged the nature or strength of the information that followed (e.g., “I don’t know much about the topic” or “It is my opinion that...”).

In training, the codings for several sample answers were discussed and prototypes of each response category were provided. Once they felt confident about these response categories, the three coders independently scored the pre-reading and post-reading responses for 12 of the 82 participants (approximately 15%). Interrater agreement (i.e., number of agreements divided by total number of information units categorized) was 0.93. All disagreements were resolved in conference with the first author, who conducted the training. The remaining questions were scored by one of the three trained coders.

3. Results and discussions

This study was designed to explore changes in readers' beliefs, perceived knowledge, demonstrated knowledge, and interest after reading two naturally occurring persuasive texts. An additional purpose was to examine differences in readers' responses to the two texts. Participants expected to be persuaded toward the authors' premises regarding prenatal similarities and the legalization of same-sex marriages after reading the articles. Significant increases in readers' perceived knowledge, demonstrated knowledge, and interest were also anticipated. However, there was an expectation that the basis for the persuasiveness of the two articles would vary to some degree. For example, the author of the Prenatal Similarities article stressed factual information to forward his arguments, whereas a more emotional response seemed more central to the persuasiveness of the Same-Sex Marriage article. Thus, knowledge should undergo more change for the Prenatal Similarities than for the Same-Sex Marriage article, resulting in a time by text interaction for perceived or demonstrated knowledge.

The means and standards deviations for participants' beliefs, perceived knowledge, demonstrated knowledge, and interest indicate differences in responses to pre-reading and post-reading measures, as well as differences in responses to the Prenatal Similarities and Same-Sex Marriage articles (see Table 1). To test the hypotheses, data were subjected to repeated measures multivariate analysis of variance (MANOVA) with beliefs, perceived knowledge, demonstrated knowledge, and interest as dependent variables, and time and text as within-subject variables. The data from these measures met all the assumptions for this procedure.

If readers were persuaded by the texts and if their perceived knowledge, demonstrated knowledge, and interest changed accordingly, there would be a significant effect for time. Further, if the paths to persuasion for the two articles differed, as hypothesized, there would be a text effect, as well as a time by text interaction. The results of the MANOVA supported these predictions. There were significant main effects for time [$F(4, 78) = 108.08, p < 0.001$] and text [$F(4, 78) = 2.58, p < 0.05$]. The time by text interaction was also statistically significant [$F(4, 78) = 5.44, p < 0.001$].

The univariate ANOVAs revealed that the omnibus effect for time was attributable to changes in readers' reported stances (i.e., beliefs) after reading the Prenatal Similarities and Same-Sex Marriage articles [$F(1,81) = 56.8, p < 0.001$,

Table 1

Means and standard deviations for beliefs, perceived knowledge, demonstrated knowledge, and interest at pre-reading and post-reading for the Prenatal Similarities and Same-Sex Marriage articles

Variable	Prenatal similarities		Same-sex marriage	
	Pre-reading <i>M</i> (<i>SD</i>)	Post-reading <i>M</i> (<i>SD</i>)	Pre-reading <i>M</i> (<i>SD</i>)	Post-reading <i>M</i> (<i>SD</i>)
Beliefs ^a	70.13(40.25)	98.60(47.25)**	94.23(47.10)	103.02(46.09)**
Perceived knowledge ^a	60.65(34.88)	86.40(30.29)**	56.20(32.66)	83.66(30.80)**
Demonstrated knowledge	2.94(4.11)	10.77(6.11)**	3.20(4.28)	9.80(7.38)**
Interest ^a	96.51(32.58)	112.04(37.91)*	92.39(41.21)	105.17(39.24)**

^a Maximum score = 150.

* $p < 0.01$.

** $p < 0.001$.

$MSe = 507.39$]. Readers' perceived knowledge, demonstrated knowledge, and interest also significantly increased after reading the two texts [$F(1,81) = 21.74$, $p < 0.001$, $MSe < 755.23$]. The main effect for text resulted from differential responses to the belief statements for the two texts, [$F(1,81) = 5.36$, $p < 0.05$, $MSe = 3111.78$], with stronger positive responses associated with the Same-Sex Marriage article than for the Prenatal Similarities article.

The time by text interaction was significant for beliefs [$F(1,81) = 15.81$, $p < 0.001$, $MSe = 501.87$] and demonstrated knowledge [$F(1,81) = 5.43$, $p < 0.05$, $MSe = 5.61$]. As shown in Fig. 1, readers expressed greater agreement with the premise of the Same-Sex Marriage article than the Prenatal Similarities article at pre-reading. However, the discrepancy in readers' stances for the two articles was less apparent at post-reading. This was due to the marked change in readers' beliefs about the similarities of humans and other mammals that resulted from reading the Prenatal Similarities article.

The more moderate change in participants' beliefs about the legalization of same-sex marriages appears to reflect the fact that their views on this issue were so highly favorable even before encountering the authors' arguments. As anticipated, participants, who were either university students or faculty, had stronger views about the legalization of same-sex marriages than they did about prenatal similarities among humans and other mammals. However, because these beliefs were generally in keeping with the authors' premise, there was limited room for persuasion to occur.

The text by time interaction for demonstrated knowledge, as depicted in Fig. 2, resulted from a greater increase in knowledge for the Prenatal Similarities article from pre-reading to post-reading than for the Same-Sex Marriage article. This pattern corresponds to our hypothesis that the Prenatal Similarities article would rely more heavily on factual content to make its case than would the Same-Sex Marriage article.

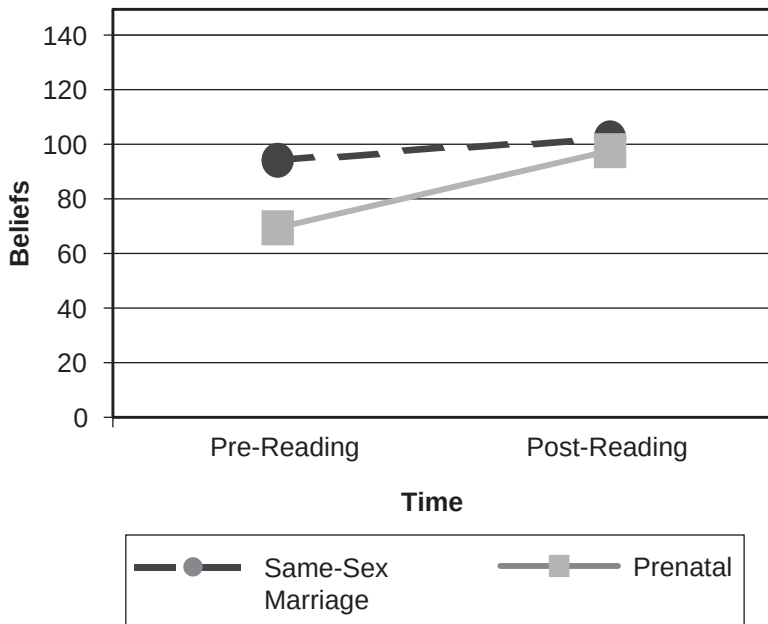


Fig. 1. Beliefs by time (pre-reading and post-reading) interaction for the Prenatal Similarities and Same-Sex Marriage articles.

3.1. Educational community and persuasion

The potential effects that undergraduate students', graduate students', and faculty's experience and training might have on their interactions with persuasive texts were also explored. Means and standard deviations for these three educational communities by text are presented in Table 2. It was hypothesized that the educational community would play a more influential role in the processing of the Prenatal Similarities article than it would for the Same-Sex Marriage article, because a number of faculty participants are nationally recognized experts in human development. On this basis, we expected that the knowledge and interests of faculty would be higher than the knowledge and interests of undergraduates and graduate students for the Prenatal Similarities article, especially at pre-reading. To test this expectation, the data for beliefs, perceived knowledge, demonstrated knowledge, and interests for each article were subjected to the repeated measures MANOVA. In this analysis, educational community was the between-subject variable and time was again the within-subject factor.

3.1.1. Prenatal similarities

There was a significant main effect for community [$F(8,152)=2.43$, $p<0.05$] and for time [$F(4,76)=79.35$, $p<0.0001$] for the Prenatal Similarities article. The univariate analyses for the Prenatal Similarities article indicated that this omnibus

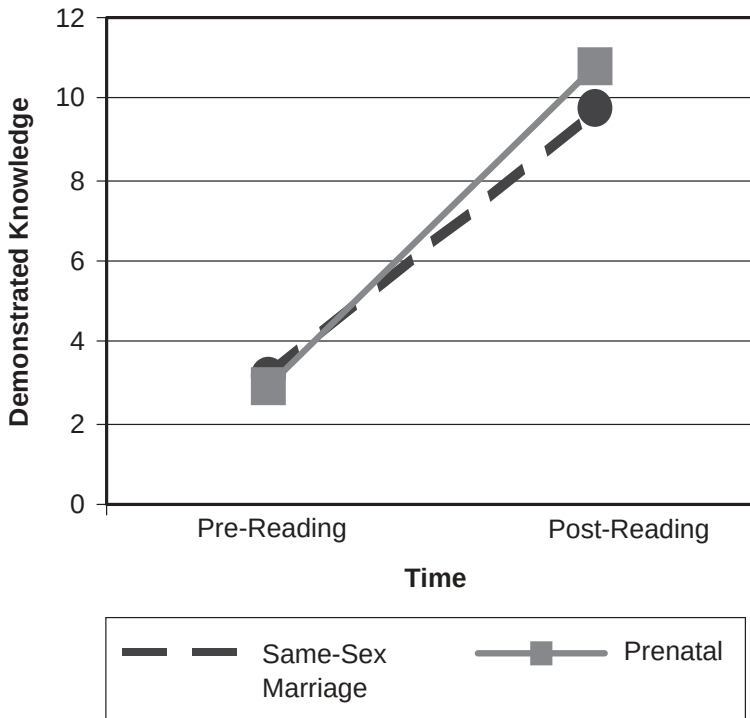


Fig. 2. Demonstrated knowledge by time (pre-reading by post-reading) interaction for the Prenatal Similarities and Same-Sex Marriage article.

effect was attributable to a significant difference among the three groups for demonstrated knowledge [$F(2,79) = 5.46$, $p < 0.01$, $MSe = 40.80$]. Post-hoc analysis was conducted using Fisher's Least Significant Differences (LSD) procedure (Fisher, 1935). This procedure has been shown to be robust in detecting significant differences among three groups (Hancock & Klockars, 1996). Post-hoc analyses indicated that faculty were the source of the significant effect for demonstrated knowledge. As expected, faculty provided significantly more information about prenatal characteristics than did either the undergraduates or graduate students both at pre-reading and post-reading.

3.1.2. Same-sex marriage

An overall effect for educational community was also found for the Same-Sex Marriage text. Specifically, the repeated measures MANOVA was significant for community [$F(8,152) = 2.79$, $p < 0.01$] and for time [$F(4,76) = 57.32$, $p < 0.0001$]. The results of univariate ANOVAs for community identified the source of the omnibus effect to be differences in beliefs for undergraduates, graduate students, and faculty [$F(2,79) = 3.34$, $p < 0.05$, $MSe = 3897.16$]. Post-hoc analysis using Fisher's LSD

Table 2

Means and standard deviations for beliefs, perceived knowledge, demonstrated knowledge, and interest at pre-reading and post-reading for the Prenatal Similarities and Same-Sex Marriage articles by educational classification

Variable educational classification	Prenatal		Same-sex marriage	
	Pre-reading <i>M</i> (<i>SD</i>)	Post-reading <i>M</i> (<i>SD</i>)	Pre-reading <i>M</i> (<i>SD</i>)	Post-reading <i>M</i> (<i>SD</i>)
<i>Beliefs^a</i>				
Undergraduates	75.14(29.36)	113.24(39.88)	79.68(48.98) _a	90.11(49.14) _a
Graduate Students	63.04(43.66)	84.48(52.06)	109.48(46.36) _b	114.92(46.19) _b
Faculty	69.75(52.44)	89.15(48.05)	102.00(37.58)	112.05(34.48)
<i>Perceived knowledge^a</i>				
Undergraduates	58.57(36.73)	88.19(28.56)	45.11(30.69)	82.16(29.12)
Graduate Students	63.88(35.69)	84.12(33.48)	60.68(34.81)	83.64(38.67)
Faculty	60.45(31.60)	85.95(30.58)	71.10(27.06)	86.45(23.17)
<i>Demonstrated knowledge</i>				
Undergraduates	1.97(2.02) _a	10.32(5.36)	2.65(2.43)	9.62(5.20)
Graduate Students	2.36(2.22) _a	8.84(3.93)	2.68(2.54)	8.16(4.30)
Faculty	5.45(7.01) _b	14.00(8.30)	4.85(7.40)	12.20(12.16)
<i>Interest^a</i>				
Undergraduates	104.43(32.22)	110.51(39.00)	90.30(41.29)	106.68(40.52)
Graduate Students	95.32(30.48)	120.40(30.94)	104.48(46.37)	110.96(41.20)
Faculty	83.35(32.83)	104.40(43.33)	81.15(31.00)	95.15(34.01)

^a Maximum score = 150.

Note: Different subscripts indicate significant differences between educational groups. For example, group means marked with the subscript b are significantly higher for a particular variable than those marked by the subscript a.

procedure indicated that graduate students held significantly stronger and more positive views about the legalization of same-sex marriages than did undergraduates both at pre-reading and at post-reading.

3.2. Knowledge, interest, and belief differences by persuasion group

The final research question in this study pertained to the perceived knowledge, demonstrated knowledge, and reported interests of individuals who were more or less persuaded by what they read. As the basis of this analysis, persuasion groups were formed using a two-step procedure. First, the differences in belief statements from pre-reading to post-reading were examined. This resulted in the formation of three tentative persuasion groups (i.e., less, moderate, and more), using ± 12 SD in belief change (in the direction of the article's premise) as our guide. The change scores for the Prenatal Similarities article ranged from –127 to 111, as compared to a range of –14 to 136 for the Same-Sex Marriage article. These data reflect the generally strong, positive views that participants held about the legalization of

same-sex marriages. They also suggest powerful, almost polar reactions some participants had to the author's argument that human embryos and fetuses are strikingly similar to those of other mammals.

Second, to make these groupings more theoretically sound, we inspected a graphic representation of the distribution as a way to confirm our natural clustering of participants. This representation was used to move certain individuals from one cluster to another and, thus, finalize our persuasion groups. As a result of these steps, 29 participants were placed in the less persuaded group, 32 in the moderate cluster, and 21 in the more persuaded group for the Prenatal Similarities article. For the Same-Sex Marriage article, 30 participants constituted the less persuaded group, 35 in the moderate cluster, and 17 made up the more persuaded group. Analyses of variance confirmed that these groups differed statistically on the basis of belief change for the Prenatal Similarities [$F(2,79) = 80.12, p < 0.0001, MSe = 534.33$] and the Same-Sex Marriage articles [$F(2,79) = 40.74, p < 0.0001, MSe = 222.28$]. Further, results of the Fisher's LSD post-hoc analyses showed that the less persuaded groups for both articles were significantly lower in belief change than the moderate groups, who, in turn, manifest statistically lower changes in beliefs than the more persuaded groups. The means and standard deviations for the dependent measures by persuasion group and by article are presented in Table 3.

With these clusters formed, we concentrated on the profiles of those in the less persuaded and more persuaded groups to understand how their beliefs, knowledge, and interest may have affected their reactions to the authors' messages. The belief was that removing the more moderate group from analysis would highlight any distinctions that might underlie the persuasion process. Next, data for each article for the more and less persuaded groups were submitted to repeated-measures MANOVAs to ascertain whether statistically significant differences in participants' knowledge and interests were indicative of belief change. Results for Prenatal Similarities article demonstrated a significant overall effect for group [$F(3,46) = 4.05, p < 0.05$] and for time [$F(3,46) = 48.21, p < 0.0001$]. The MANOVA for the Same-Sex Marriage article was significant only for time, however [$F(3,43) = 47.45, p < 0.0001$]. Therefore, subsequent discussion will focus only on the Prenatal Similarities article.

The univariate ANOVAs for the Prenatal Similarities showed that the source of the difference between the more or less persuaded groups reflected differences in their perceived knowledge [$F(1,48) = 9.84, p < 0.01, MSe = 1389.43$]. In short, those who were more persuaded by the authors' message had significantly lower perceived knowledge scores at pre-reading than those who were less swayed by the text arguments. It might be argued that higher ratings of perceived knowledge might have reduced the less-persuaded group's desire to process or elaborate the arguments (Cacioppo & Petty, 1982; Dole & Sinatra, 1998), contributing to their relatively intractable beliefs about prenatal similarities among mammals.

3.3. *The interplay between beliefs, knowledge, and interests*

As an additional means of exploring the interplay of person and message in the persuasion process, correlation matrices for the Prenatal Similarities (Table 4) and

Table 3

Means and standard deviations for beliefs, perceived knowledge, demonstrated knowledge, and interest at pre-reading and post-reading for the Prenatal Similarities and Same-Sex Marriage articles by persuasion group

Variable persuasion level	Prenatal		Same-sex marriage	
	Pre-reading <i>M</i> (<i>SD</i>)	Post-reading <i>M</i> (<i>SD</i>)	Pre-reading <i>M</i> (<i>SD</i>)	Post-reading <i>M</i> (<i>SD</i>)
<i>Beliefs^a</i>				
Less	74.62(50.94) _b	64.48(57.65)	107.17(48.46) _b	104.30(49.66)
Moderate	80.47(31.10)	114.81(28.79)	91.86(49.39)	97.03(49.42)
More	48.19(26.89) _a	121.00(21.68)	76.29(33.53) _a	113.12(30.05)
<i>Perceived knowledge^a</i>				
Less	70.34(33.09) _b	91.17(30.23)	53.17(34.30)	83.57(33.45)
Moderate	62.31(37.67)	92.66(29.75)	63.89(32.84)	84.89(32.35)
More	44.71(28.05) _a	70.29(26.36)	45.71(26.69)	81.29(23.31)
<i>Demonstrated knowledge</i>				
Less	3.24(4.21)	10.55(6.87)	3.63(5.77)	10.23(10.45)
Moderate	2.37(2.54)	11.31(5.21)	3.09(3.44)	9.06(4.99)
More	3.38(5.71)	10.24(6.50)	2.65(2.52)	10.59(4.74)
<i>Interest^a</i>				
Less	96.38(28.48)	108.00(39.56)	98.30(43.38)	109.83(36.82)
Moderate	94.88(36.03)	116.47(39.17)	87.17(43.79)	101.63(42.63)
More	99.19(33.76)	110.86(34.56)	92.71(31.42)	104.24(37.53)

^a Maximum score = 150.

Note: Different subscripts indicate significant differences between the less or more persuaded groups. For example, group means marked with the subscript *b* are significantly higher for a particular variable than those marked by the subscript *a*.

Same-Sex Marriage articles (Table 5) were examined. This enabled an investigation of the associations between participants' beliefs and their perceived and demonstrated knowledge, as well as their reported interest in the article topic. Of special interest were the similarities and differences in the correlational patterns for the two compelling, but varied, texts.

For one, the two forms of knowledge assessed in this study, perceived and demonstrated, were not significantly related either at pre-reading or at post-reading for either article. This reinforces the need to consider multiple forms of knowledge in the study of persuasion. Further, these correlation data indicated that these two forms of knowledge were differentially linked to individuals' beliefs and interests. Demonstrated knowledge, for example, did not significantly relate to readers' beliefs for either article, whereas perceived knowledge did significantly relate to beliefs.

However, the nature of this relationship between perceived knowledge and beliefs varied by article. For the Prenatal Similarities article, readers' initial beliefs were positively linked to their level of perceived knowledge after reading. Interestingly,

Table 4

Correlations for beliefs, perceived knowledge, demonstrated knowledge, and interests for the Prenatal Similarities article

	1 Pre- beliefs	2 Post- beliefs	3 Pre- perceived knowledge	4 Post- perceived knowledge	5 Pre- demonstrated knowledge	6 Post- demonstrated knowledge	7 Pre- interest	8 Post- interest
1	1.000	0.598**	−0.050	0.267*	−0.189	−0.102	−0.111	0.098
2		1.000	−0.238*	−0.018	−0.141	−0.049	−0.074	0.107
3			1.000	0.575**	−0.005	−0.068	0.242*	0.291**
4				1.000	−0.141	−0.058	0.261*	0.264*
5					1.000	0.724**	0.096	−0.299**
6						1.000	0.103	−0.125
7							1.000	0.182
8								1.000

* $p < 0.05$.** $p < 0.01$.

Table 5

Correlations for beliefs, perceived knowledge, demonstrated knowledge, and interests for the Same-Sex Marriage article

	1 Pre- beliefs	2 Post- beliefs	3 Pre- perceived knowledge	4 Post- perceived knowledge	5 Pre- demonstrated knowledge	6 Post- demonstrated knowledge	7 Pre- interest	8 Post- interest
1	1.000	0.899**	0.279*	0.283**	0.051	−0.015	0.492**	0.514**
2		1.000	0.320**	0.337**	0.001	−0.049	0.486**	0.422**
3			1.000	0.595**	0.047	−0.112	0.353**	0.298**
4				1.000	0.055	−0.042	0.186	0.250*
5					1.000	0.858**	−0.031	0.135
6						1.000	0.022	0.205
7							1.000	0.756**
8								1.000

* $p < 0.05$.** $p < 0.01$.

there was a significant negative relationship between readers' pre-reading perceived knowledge and their post-reading beliefs about the prenatal similarities among humans and other animals. This means that readers with higher levels of initial perceived knowledge tended to express less agreement with the author after reading the article. Perhaps readers' stronger perceptions of their knowledge promoted a more critical analysis of the authors' arguments about such prenatal characteristics.

For the Same-Sex Marriage article, perceived knowledge and beliefs were significantly positively correlated at pre-reading and post-reading. This pattern likely reflects the fact that the issue of same-sex marriages plays more to pathos, in

Aristotle's terms, than to ethos. Further, the authors depended on the personal story of a gay couple more than factual content to forward their position. Therefore, readers' initial knowledge perceptions were not as apt to be challenged by the article's content, as was the case in the Prenatal Similarities text.

Differences in the correlational patterns for the two articles were found for other variables as well. Readers' beliefs about prenatal similarities were not significantly related to their interest in the topic, but there were relatively strong correlations between readers' beliefs and interest in the legalization of same-sex marriages. Also, pre-reading demonstrated knowledge was negatively related to post-reading interest for the Prenatal Similarities article, indicating that readers with higher levels of knowledge before reading the article had lower levels of interest in the topic after reading. However, demonstrated knowledge and interest were not related for the Same-Sex Marriage article. Interestingly, the correlations between perceived knowledge and interest were positive for both articles. This suggests that what readers feel they know about a topic may play into their interests in that subject more than what they actually know.

These correlational findings support the results of the other analyses. First, they indicate that the variables of perceived and demonstrated knowledge may contribute differently to the persuasion process. Thus, those developing models of persuasion and conceptual change may wish to consider both of these dimensions of knowledge. Second, the differences in the interrelationships among the variables for the two persuasive texts provide further evidence that the paths to persuasion may differ, particularly when a topic is cast more emotionally than factually. Further analysis of this issue seems warranted.

4. Conclusions and implications

The literature on persuasion and conceptual change has identified key person and text factors that instigate changes in those hearing or reading messages intended to evoke such change. The goal of this study was to extend that research by considering conditions others have judged important but difficult to assess in the persuasion process (Dole & Sinatra, 1998; Petty & Cacioppo, 1986). These conditions include the specificity of the constructs; the interplay among variables; the influence of processing non-contrived, controversial texts; intra-individual response patterns; and the impact of respondents' educational backgrounds. The outcomes reported herein set the stage for subsequent studies of persuasion by shedding light on the person and text factors underlying changes in reader understanding and beliefs.

Of course, the contributions of this study to the persuasion literature must be understood in terms of its potential limitations. First, the persuasion process was investigated using naturally occurring texts, rather than materials expressly contrived for experimental purposes. As a result, particular textual features that may have heightened the statistical comparability of our persuasive articles could not be controlled, as other researchers have done (Kardash & Scholes, 1996). Consequently, it was impossible to tease apart the relative effects of certain textual

elements, such as topic, clarity of arguments, or author credibility, as has been done in prior studies (Chambliss, 1995; Slusher & Anderson, 1996). Moreover, the sample consisted of mature readers who were members of three educational communities. Outcomes that may arise when participants are less competent readers or less educationally oriented remain to be seen.

In addition, because much of our prior research has centered on knowledge, interest, and beliefs (e.g., Alexander et al., 1995; Alexander et al., 1998), those same components were included in this study. It can be argued that other variables or alternative measures of the included variables might paint a different picture of the persuasion process. However, the components targeted are elements of various theories and models of belief change (e.g., Dole & Sinatra, 1998; Murphy, 1998) and have been incorporated in empirical investigations of persuasion and conceptual change (e.g., Anderson & Sechler, 1986; Wood & Eagly, 1981). What has not been considered in this multifaceted study is any predisposition to argumentation, such as open-mindedness or resistance, that may well exist (Stanovich & West, 1997). Nor was any specific measure of emotional reaction to the article included (Murphy, 1998). Such interesting prospects must be considered in subsequent research. Even in light of these limitations, there are consequential outcomes from this study that provide clues as to why and how persuasion occurs.

4.1. Particular measures of general constructs

In forwarding their model of cognitive change, Dole and Sinatra (1998) state that one of the challenges for those concerned with persuasion and conceptual change is identifying and assessing the impact that varying types of knowledge play in these processes. This investigation included two forms of knowledge that on the surface appear rather similar, perceived knowledge (i.e., what one thinks one knows) and demonstrated knowledge (i.e., the relevant knowledge one produces). These two constructs operated differently in the persuasion process. Our findings suggest that perceptions of knowledge, as with research on self-efficacy (e.g., Zimmerman, Bandura, & Martinez-Pons, 1992), are more potent forces in how one approaches a controversial topic than what one may actually know. It would also appear that those with expertise are more conservative in their judgments than others, perhaps because of their awareness of the breadth and depth of the domain. Further, even when judgments of knowledge seem misguided, they pertain to one's interest in the topic. Somewhat tempered levels of perceived knowledge may, in effect, signal the "need to know," that is so important to subsequent changes in attitudes (Cacioppo & Petty, 1982) and conceptual understanding (Dole & Sinatra, 1998).

Although perceived and demonstrated knowledge were expected to operate somewhat differently in the persuasion process, it was not assumed that their relationship would be statistically non-significant. This anomaly can be attributed to the fact that our experts, who demonstrated the most knowledge of both topics at pre-reading and post-reading, conservatively judged their knowledge base. Even in the area of their research (i.e., human development), experts' knowledge ratings were comparable to the ratings of undergraduates and graduate students. Also, the

open-ended nature of our demonstrated knowledge task understandably captured only a very select portion of participants' understanding, especially for the experts. For this reason, we are currently experimenting with a demonstrated knowledge measure that incorporates both open and closed questions, as a means of examining both the breadth and depth of participants' demonstrated knowledge and its subsequent relation to perceived knowledge.

4.2. Interplay of person variables

As models of change, the ELM and the CRKM consider how the variables of knowledge and interest contribute to shifts in attitude or conceptual understanding. However, the specific nature of the presumed relationship between knowledge and interest in these leading models was not ascertained. That is, it was not apparent whether knowledge and interest should be strongly correlated or whether lower or higher levels of interest would limit change. Although knowledge and interest were significantly correlated in this investigation, the clear association we have typically found for these two variables in our research (e.g., Alexander et al., 1995) did not emerge here.

This pattern can be attributed largely to the experts in this study who consistently had the highest means for demonstrated knowledge, but the lowest means for topic interest. Perceived knowledge and reported interest were far better aligned for these expert participants. Truly, the experts in this study offered a unique portrait of the persuasion process, as compared to the undergraduates and graduate students who participated. These experts remind us that stimulating the interest, enhancing the knowledge, and changing the beliefs of those who are immersed in a relevant field is no easy undertaking (Petty & Cacioppo, 1986). Our findings do indicate that the interplay between knowledge and interest is quite complex and not as straightforward as existing models may portray, especially when alternate forms of knowledge are considered.

4.3. Nature of the controversial topic

Models of persuasion and conceptual change do give credence to text factors as contributors to the change process. However, there is more to text than whether arguments are well made or the writing is cohesive. There is also the nature of the issue being discussed and whether that issue may be more readily addressed via knowledge-based arguments or by playing to the readers' emotions. Thus, when Kardash and Scholes (1995) focus on AIDS, and specifically on readers' scientific misconceptions regarding this dreadful disease (e.g., AIDS can be transmitted via mosquito bites or casual contact), they are potentially researching something quite different from those who seek to convince their audience that AIDS is not a divine plague or retribution for ungodly behavior.

Our interest in belief change in this investigation is more in keeping with the latter than the former in that we have posed controversial issues to mature readers for which obviously "correct" or "accurate" positions do not exist. This was especially

true for the topic regarding the legalization of same-sex marriages. When the focus of learning is on controversial topics for which specific factual content is insufficient to alter views (Chinn & Brewer, 1993; Vosniadou & Brewer, 1987), persuasion, or the change in individuals' beliefs relative to the topic, cannot be cast in an accurate or inaccurate or all-or-nothing way. In effect, in this situation, individuals are apt to be more or less susceptible to the author's arguments and, therefore, more or less accepting of the guiding premise (Miller, 1980).

In this study, the knowledge path was more pronounced for the Prenatal Similarities article and the changes in beliefs were more moderate for the text dealing with the legalization of same-sex marriages. While tentative, such results suggest the value of looking more deeply at the nature of the topic or domain to be processed. This point is significant because of rising interest in conceptual change and persuasion in the literature, and because of the expanding call for more collaboration between researchers immersed in these two related domains (Dole & Sinatra, 1998; Murphy, 1998). As Chinn (1999) has suggested, the lack of attention to topic or domain in these related fields has served to thwart such collaboration. Thus, if this laudable collaboration is to prove fruitful, the nature of the change under consideration and the conceptions of knowledge and beliefs that researchers in these fields hold must be brought to the forefront.

Before this discussion is concluded, it is important to reassert the value of this line of inquiry. As stated at the outset, the study of persuasion has a long and distinguished history. Ancients of the likes of Plato and Aristotle well understood the power persuasion held for human learning and development (Carey, 1994; Cooper, 1932). Persuasion is no less a potent force in today's post-industrial societies. Indeed, when the frequency with which individuals in this information age encounter oral or written messages constructed for the explicit or implicit purpose of altering their fundamental views is considered, it seems imperative to proceed with our quest to understand the very nature of the persuasive process. Clearly, this study reinforced the notion that persuasion, even among skilled and knowledgeable readers, *does* occur. Further, this investigation serves as an initial step toward answering the question of *why* and *how* persuasion occurs. As such, it contributes substantively to our understanding of the interplay of person and message that makes persuasion a more or less likely outcome, and, thus, plants the seeds for future explorations into the dynamic nature of persuasion.

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